DSC 40B - Discussion 07

First let's review problems that a lot of students had trouble with on the midterm.

Problem 1.

Consider the function $f(n) = \ln(n \cdot 2^n) - 50\sqrt{n}(\log n - 3)$. Which of the following asymptotic bounds are true? Choose **all** that apply.

- ○ $\Theta(n^2)$
- ● $O(n^2)$
- ○ $\Theta(n \log n)$
- ● $\Theta(n)$
- ● $O(n)$
- ○ $O(\sqrt{n} \log n)$
- ● $\Omega(1)$

Problem 2.

Consider this version of `mergesort`, which is the same as the one we saw in lecture but with an added `print` statement:

```
import math
def mergesort(arr):
    """Sort array in-place."""
    print("Here!")
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

Suppose this is called on an array of size 8. How many times will `"Here!"` be printed?

15

Solution:

see midterm solution

Problem 3.

Suppose `bAA` and `bBB` are two functions. Suppose `bAA`'s time complexity is $\Theta(n^2)$, while `bBB`'s time complexity is $\Theta(\sqrt{n})$.

Suppose `foo` is defined as below:

```python
def foo(n):
    if n < 1000:
        for i in range(n):
            bAA(n)
    else:
        for j in range(n):
            bBB(n)
```

What is the asymptotic time complexity of `foo`?

$\Theta(n\sqrt{n})$

Problem 4.

What is the **expected** time complexity of the following function? State your answer using asymptotic notation.

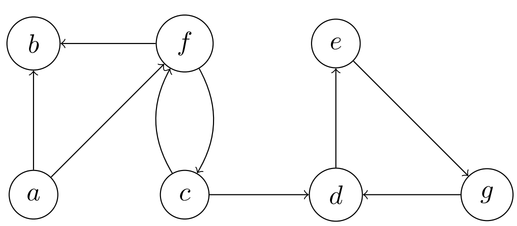
```python
def foo(n):
    # draw a number uniformly at random from 0, 1, 2, ..., n-1 in Theta(1)
    x = random.randrange(n)

    if x > n - math.sqrt(n):
        for j in range(n**2):
            print(j)
    else:
        for j in range(n):
            print(j)
```

$\Theta(n^{1.5})$

Problem 5.

Consider a *breadth*-first search on the graph shown in the figure, starting with node *c*.



a) Suppose you call bfs_shortest_paths(graph, 'c') on the graph above. This function returns dictionaries distance and predecessor. Write down the contents of these dictionaries as they are when the function exits.

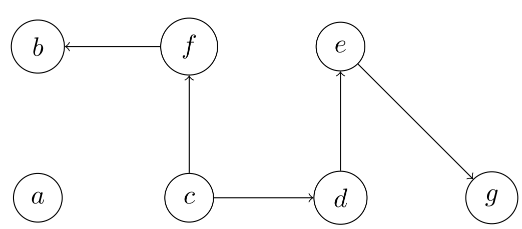
Solution:
distance: {
'a':∞,
'b':2,
'c':0,
'd':1,

```python
def bfs_shortest_paths(graph, source):
    status = {node:'undiscovered' for node in graph.nodes}
    distance= {node:float('inf') for node in graph.nodes}
    predecessor = {node: None for node in graph.nodes}
    status[source] = 'pending'
    distance[source]=0
    pending = deque([source])
    # while there are still pending nodes
    while pending:
        u = pending.popleft()
        for v in graph.neighbors(u):
            # explore edge (u,v)
            if status[v] == 'undiscovered':
                status[v]='pending'
                distance[v]=distance[u]+1
                predecessor[v]=u
                # append to right
                pending.append(v)
        status[u]='visited'
    return predecessor, distance
```

’e’:2,
’f’:1,
’g’:3
}
predecessor:{
’a’:None,
’b’:’f’,
’c’:None,
’d’:’c’,
’e’:’d’,
’f’:’c’,
’g’:’e’
}

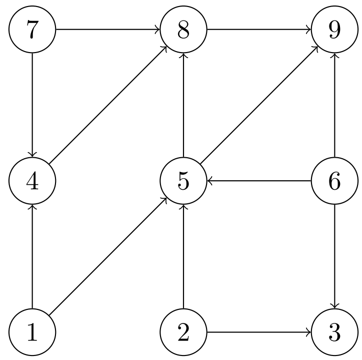
b) Mark the BFS trees produced on executing BFS on this graph.

Solution:



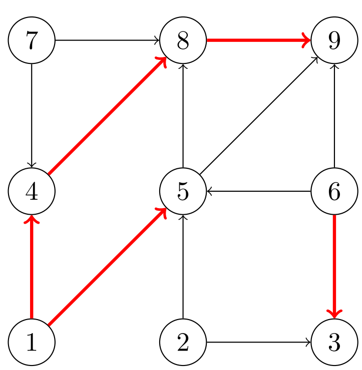
Problem 6.

Consider the following directed graph.



a) Run Full_DFS on the graph above. Make a bold arrow from node u to node v if u is the predecessor of node v in DFS. Use the convention that a node's neighbors are processed in ascending order by label.

Solution:



b) Fill in the table below so that it contains the start and finish times of each node after a Full_DFS is performed on the above graph. Assume node 1 as the source for the first DFS call. Begin your start times with 1.

Node	Start	Finish
1	1	10
2	17	18
3	14	15
4	2	7
5	8	9
6	13	16
7	11	12
8	3	6
9	4	5

c) Topologically sort the vertices of the graph.

Solution: 2,6,3,7,1,5,4,8,9

Problem 7.

State whether the following statements are true or false.

a) Full breadth first search on a directed graph always produces same number of BFS trees irrespective of order in which vertices are given and the neighbouring nodes are visited.

Solution: False
For a directed graph, the number of BFS trees can vary depending on the order in which vertices are given.

b) Full breadth first search on an undirected graph always produces same number of BFS trees irrespective of order in which vertices are given and the neighbouring nodes are visited.

Solution: True.

c) Both Full BFS and Full DFS require atleast $\Omega(V)$ memory.

Solution: True
Both Full BFS and Full DFS require $\Omega(V)$ memory to store the status of the nodes.

d) Consider a graph G on which BFS is run with node s as the source, and $d(a, b)$ is the distance from node a to b. Assume that BFS visits a node u in the graph before node v. Then $d(s, u) < d(s, v)$.

Solution: False. $d(s, u) \leq d(s, v)$

e) Every directed acyclic graph has exactly one topological ordering.

Solution: False. There can be multiple topological orderings depending on the order of nodes chosen during DFS.

Problem 8.

Given an undirected graph G=(V,E), give an algorithm to find if the graph is disconnected.

Solution: A connected graph is a graph with one connected component, this means the only CC must have $|V|$ nodes.

Note that running BFS or DFS on a node in an undirected graph will visit all nodes in the connected component containing that node.

Thus, we can run BFS or DFS on a node in G (since G is undirected, any node will do). If the size of the set of visited nodes does not equal $|V|$, then the graph is disconnected.

Figure 1: "Full" DFS

```
from data classes import dataclass
    @dataclass
    class Times:
        clock: int
        start: dict
        finish: dict

    def full_dfs_times(graph):
        status = {node:'undiscovered' for node in graph.nodes}
        predecessor = {node: None for node in graph.nodes}
        times=Times(clock=0, start={}, finish={})
        for u in graph.nodes:
            if status[u]=='undiscovered':
                dfs_times(graph, u, status, times)
        return times, predecessor

        def dfs_times(graph, u, status, predecessor, times):
            times.clock+=1
            times.start[u]=times.clock
            status[u]='pending'
            for v in graph.neighbors(u):
                # explore edge (u, v)
                if status[v]=='undiscovered':
                    predecessor[v]= u
                    dfs_times(graph, v, status, times)
            status[u]='visited'
            times.clock+=1
            times.finish[u]=times.clock
```